

Nginx Reverse Proxy

▼ Relevant source files

Backend/STUDER/Dockerfile

Backend/STUDER/pom.xml

Docker/docker-compose.yml

Docker/nginx.conf

The Nginx reverse proxy serves as the entry point for the STUDER ecosystem. It handles traffic routing between the Angular frontend and the Spring Boot backend, manages WebSocket upgrades for real-time features, and enforces global constraints such as payload size limits. By abstracting the internal network topology, it allows the system to present a unified interface to the user while maintaining isolation between the `frontend_net` and `backend_net` Docker networks.

Proxy Architecture and Data Flow

The configuration defines two primary server blocks. The main listener on port `1583` handles standard application traffic, while a secondary listener on port `5712` provides an alternative access point for backend-specific routing.

Traffic Routing Logic

1. **API Traffic:** Any request starting with `/api/` is forwarded to the `app_upstream` group, which resolves to the `app-be` container on port `8081` `Docker/nginx.conf` `24-26`

`Docker/nginx.conf` `33-39`

2. **Frontend Traffic:** All other requests (root `/`) are directed to the `frontend_upstream`, which resolves to the `frontend` container on port `80` `Docker/nginx.conf` `18-20`

`Docker/nginx.conf` `43-50`

Request Routing Flow

Sources: `Docker/nginx.conf` `18-51` `Docker/docker-compose.yml` `59-72`

Configuration Details

Global Settings

The proxy is configured to handle high-concurrency and large media uploads:

- Worker Connections:** Set to `1024` per process `Docker/nginx.conf` `8`
- Payload Limit:** `client_max_body_size` is set to `20m` to accommodate image uploads (e.g., profile pictures or message attachments) before they are processed by the media service `Docker/nginx.conf` `16`
- Keepalive:** Optimized at `65` seconds for persistent connections `Docker/nginx.conf` `15`

Upstream Definitions

The configuration uses named upstreams to abstract the service discovery provided by Docker's internal DNS:

Upstream Name	Destination	Container Name
<code>frontend_upstream</code>	<code>frontend:80</code>	<code>studer_frontend</code>
<code>app_upstream</code>	<code>app-be:8081</code>	<code>studer_app_be</code>

Sources: `Docker/nginx.conf` `18-26` `Docker/docker-compose.yml` `17-18`
`Docker/docker-compose.yml` `51-52`

Header Management and WebSockets

The proxy ensures that client metadata is preserved during the hop and supports protocol switching for WebSockets:

- X-Forwarded Headers:** Injects `X-Real-IP`, `X-Forwarded-For`, and `X-Forwarded-Proto` so the Spring Boot backend can identify the original requester `Docker/nginx.conf` `35-38`
- WebSocket Support:** The root location includes `Upgrade` and `Connection` headers, enabling the Angular frontend to maintain persistent connections if required by the dev server or future features `Docker/nginx.conf` `45-47`

Service Topology

Nginx bridges two distinct bridge networks defined in the `docker-compose.yml`. This ensures that while the frontend cannot directly access the database or MinIO, the Nginx container can reach

both the user-facing frontend and the internal backend services.

Network Bridge Mapping

Sources: Docker/docker-compose.yml 90-94 Docker/docker-compose.yml 59-72
Docker/nginx.conf 28-75

Secondary Server Block (Port 5712)

A secondary server block is defined on port 5712 with the server_name set to backend Docker/nginx.conf 53-55 While it mirrors the routing logic of the main port (routing /api/ to app_upstream and / to frontend_upstream), it serves as a dedicated listener for internal administrative routing or secondary access paths within the containerized environment Docker/nginx.conf 59-74

Implementation Summary Table

Feature	Configuration Setting	Purpose
External Port	listen 1583	Primary entry point for users Docker/nginx.conf 29
Backend Port	listen 5712	Secondary/Backend-specific listener Docker/nginx.conf 54
Max Upload	client_max_body_size 20m	Allows multipart image uploads Docker/nginx.conf 16
API Proxy	location /api/	Routes to Spring Boot app-be Docker/nginx.conf 33
Frontend Proxy	location /	Routes to Angular frontend Docker/nginx.conf 43
WS Support	proxy_set_header Upgrade	Enables WebSocket protocol switching Docker/nginx.conf 46

Sources: Docker/nginx.conf 1-76